

Household Appliance Energy Forecasting: A Comparative Time-Series Case Study

Dataset: UCI Appliance Energy Prediction (Candanedo, Feldheim & Deramaix, 2017)

Link to Github: <https://github.com/AjayiIsrael/Appliance-Energy-Forecasting>

Abstract

This report forecasts hourly household appliance energy use 24 hours ahead using the UCI Appliance Energy Prediction dataset (Candanedo, Feldheim & Deramaix, 2017), resampled to hourly resolution (3,290 observations). Five naive benchmarks, SARIMAX (AIC-searched order), two gradient-boosted models (XGBoost, LightGBM), and the Chronos-Bolt-small zero-shot foundation model are compared on an identical 14-day rolling-origin backtest. The two strongest models are effectively tied: SARIMAX achieves the best RMSE (325.8) while Chronos-Bolt-small — fit with no dataset-specific tuning at all — achieves the best MASE (0.638), narrowly ahead of SARIMAX (0.656); both clearly outperform the engineered-feature ML models and every naive benchmark. A leakage experiment shows using true future weather (rather than genuinely available lagged weather) inflates apparent accuracy by 13.7% RMSE, and a feature ablation shows weather covariates add no value, and mildly hurt LightGBM, once lag and time-of-day features are present. SARIMAX is recommended as the practical choice for a single-house deployment; Chronos-Bolt-small is preferable at multi-household scale, where its zero-shot property removes per-house tuning entirely.

1. Introduction, Dataset and Preprocessing

Forecasting household appliance demand a day ahead supports demand-response scheduling, battery/storage decisions, and grid-load planning. This report compares five model families — naive benchmarks, SARIMAX, two gradient-boosted trees, and a pretrained foundation model — on an identical backtest, so differences in accuracy reflect the model rather than the test split.

The dataset (Candanedo, Feldheim & Deramaix, 2017) records appliance energy use (Wh), indoor temperature/humidity in nine rooms, and outdoor weather (temperature, humidity, pressure, wind speed, visibility, dew point) at 10-minute resolution in a low-energy house in Stambruges, Belgium, 11 Jan–27 May 2016 (19,735 rows). The data is complete (zero missing values, zero timestamp gaps) and was retrieved programmatically from the UCI repository, with an automatic fallback to a GitHub mirror if the primary host is unreachable. Two random-noise control columns (rv1, rv2), included by the original authors only to test feature-selection methods, were dropped.

For this analysis the series was resampled to hourly resolution (3,290 observations): energy columns (Appliances, lights) were summed within each hour, since Wh is additive, while sensor and weather columns were averaged, since they are instantaneous measurements rather than accumulations.

2. Forecasting Problem and Evaluation Design

Target: Appliances (hourly Wh). Forecast 24 hours ahead. Each model, is checked using the exact same 14-day rolling-origin backtest. That is, 14 separate forecasts at 24-hour-ahead horizon, one per day, and for each one the model is trained only on observations strictly before its own origin point. We rejected a single fixed train/test split, because the timeline here ends shortly after a rather unusually large demand spike, and that would skew a one-window evaluation toward the model that reacted best to that one event. The 14 origins, taken together, span exactly the final $14 \times 24 = 336$ hours of the series, so this design also satisfies a strict chronological split over the last 14 days.

Four metrics are reported: MAE and RMSE (Wh, directly interpretable), sMAPE (bounded percentage error), and MASE (Mean Absolute Scaled Error, scaled against in-sample seasonal-naive MAE) — the primary metric, since it is scale-free and interpretable relative to a trivial benchmark ($\text{MASE} < 1$ beats seasonal persistence).

3. Exploratory Analysis and Stationarity

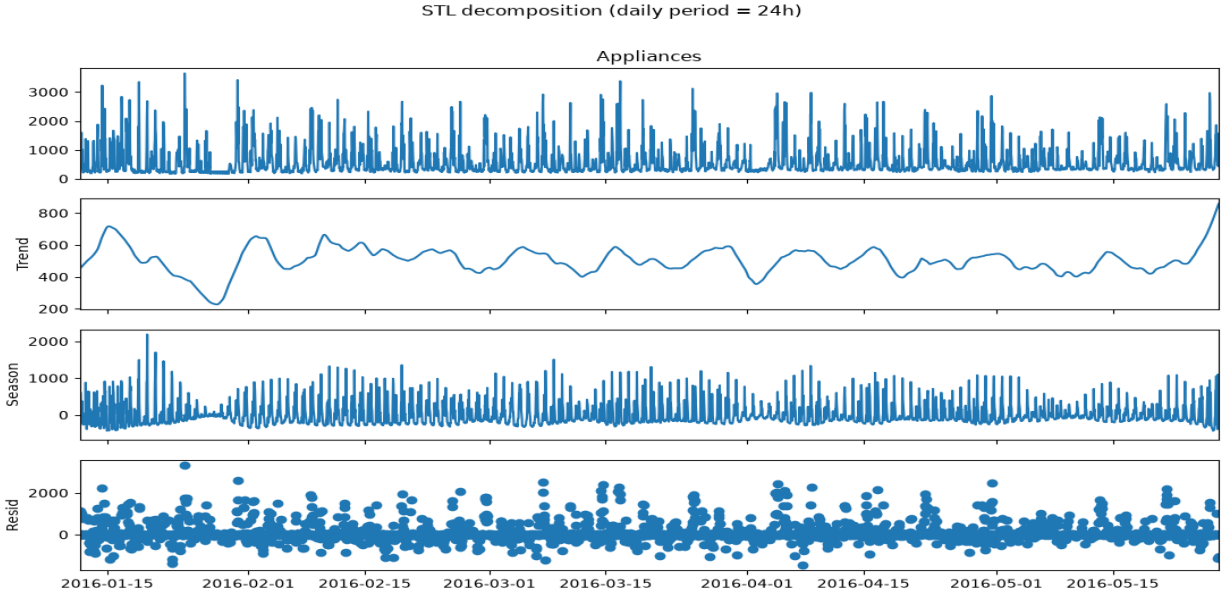


Figure 1. STL decomposition (24h period): raw series, trend, seasonal, and residual components.

STL decomposition (Figure 1) shows a moderate daily seasonal component (strength 0.318), a slightly stronger weekly component (0.368, weekday/weekend differences), and the trend is basically absent (0.048) over the 4.5-month window. Two stationarity tests with opposite null hypotheses agree the raw hourly series is stationary in levels: ADF (H_0 : unit root) rejects non-stationarity strongly (statistic = -9.127 , $p = 3.1 \times 10^{-15}$), and KPSS (H_0 : stationary) fails to reject stationarity (statistic = 0.042, $p = 0.10$). This directly justifies $d = 0$ in the SARIMAX order search (Section 5).

4. Benchmark Models

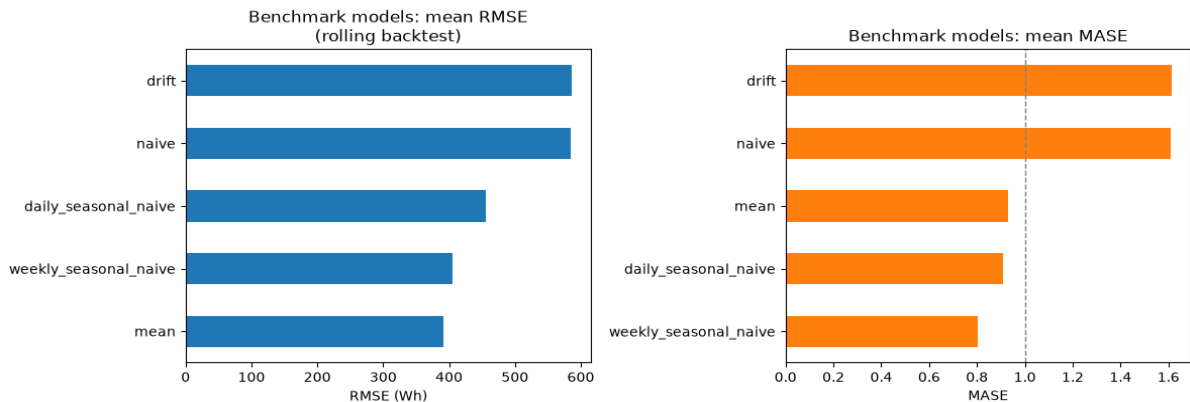


Figure 2. Benchmark models: mean RMSE and MASE, 14-origin backtest.

Five parameter-free forecasts anchor the comparison: historical mean, last-value persistence (naive), 24h and 168h seasonal-naive, and linear drift. Weekly seasonal-naive is strongest (MASE 0.802), ahead of the mean (0.932) and daily seasonal-naive (0.908); naive and drift are far worse (MASE 1.608 and 1.613) since the series

has no material trend to extrapolate and last-hour persistence ignores its daily/weekly structure. That weekly beats daily seasonal-naïve indicates day-of-week effects carry real forecasting value beyond time-of-day alone.

5. SARIMAX

SARIMAX (Seasonal ARIMA with eXogenous regressors, no exogenous inputs used) suits this series' strong, regular, stationary seasonality directly, without feature engineering. The non-seasonal order was chosen by an exhaustive AIC grid search over $p \in [0,6]$, $d \in [0,2]$, $q \in [0,6]$ (147 candidates, fit on the last 45 days for tractability): the search selected (0,0,6) among converged fits (AIC = 15,143.9), consistent with $d = 0$ from Section 3. The seasonal order was then chosen by comparing four candidates at (P,D,Q,24): (2,0,1,24) was clearly best (AIC = 14,895.8) over (1,0,1,24) at 15,143.9, confirming the seasonal terms do real work. The final SARIMAX(0,0,6)(2,0,1,24) was refit on the full training series (AIC = 47,149.2, BIC = 47,209.9).

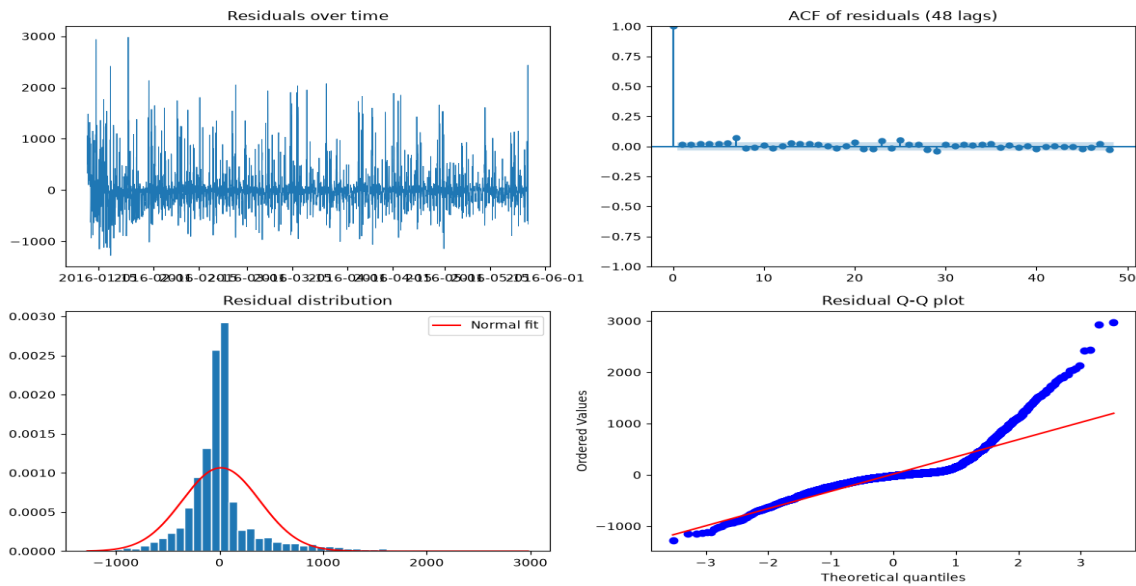


Figure 3. SARIMAX residual diagnostics: residuals over time, residual ACF, distribution vs. normal fit, Q-Q plot.

Residual diagnostics (Figure 3) show the model has not captured all structure: Ljung-Box rejects no-autocorrelation at lags 24 and 48 ($p = 0.0018, 0.0023$) — plausibly the weekly effect, since the seasonal period was restricted to $m=24$ for tractability — and residuals are right-skewed (2.01) and heavy-tailed (kurtosis 8.82), consistent with the occasional large demand spikes visible in Figure 1: a linear Gaussian model systematically under-predicts rare large spikes and over-predicts around them.

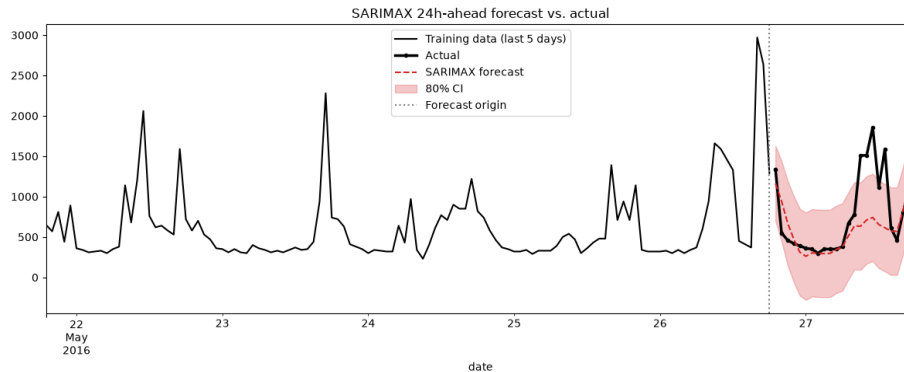


Figure 4. SARIMAX 24h-ahead forecast vs. actual, 80% CI, for the final backtest origin.

Despite the residual issues, SARIMAX's 14-origin backtest achieves the best RMSE of any model (MAE 208.3, RMSE 325.8, sMAPE 29.0%, MASE 0.656); per-origin RMSE ranged from 141.0 (calmest day) to 831.2 (most volatile day), showing the dataset's difficulty is driven by a handful of large events rather than steady day-to-day noise.

6. Covariates and the Data-Leakage Experiment

Indoor sensor, outdoor weather, and time-of-day/day-of-week features were engineered as candidate covariates for the ML models (Section 7). Weather/sensor columns come in two variants: realistic (lagged 24h, genuinely available at forecast time, since tomorrow's actual weather is not known today) and conditional (true future values, an oracle upper bound that is a common and easy mistake to introduce when engineering features from a static historical file where every column already exists for every row).

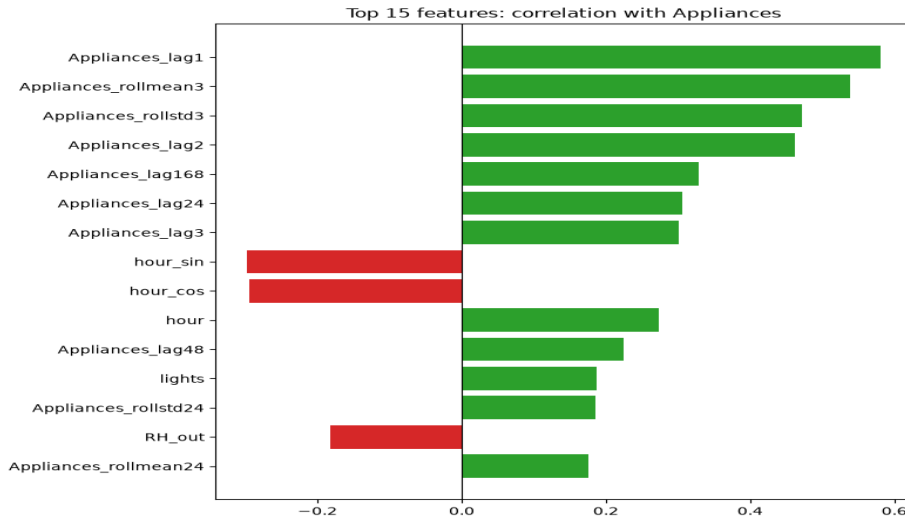


Figure 5. Top 15 engineered features by absolute correlation with the target.

Correlation with the target (Figure 5) is dominated by the target's own history — lag-1 ($r=0.581$), a 3-hour rolling mean ($r=0.538$), rolling std ($r=0.473$) and lag-2 ($r=0.463$) are the strongest predictors by a wide margin, with the weekly lag ($r=0.329$) ahead of the daily lag ($r=0.306$), echoing Section 3's finding that weekly seasonality is at least as strong as daily. Only one weather variable, outdoor humidity (RH_out, $r=-0.182$), reaches the top 15 at all — an early signal that weather contributes little once the target's own recent behaviour is available, confirmed directly by the ablation in Section 7.

Quantifying the leakage: the same LightGBM model fit on the final test window with realistic features reached RMSE = 345.4, versus RMSE = 303.9 with conditional (oracle) features — a 13.7% RMSE cost of using only genuinely available information. This is the measured price of an honest forecast: any pipeline using same-hour weather as a same-hour predictor without lagging it is silently reporting the easier, conditional number.

7. Feature-Based Machine Learning Model

XGBoost and LightGBM were trained on the realistic feature set (lags 1–2–3–24–48–168h; rolling mean/std 3–24–168h; cyclically-encoded hour/day-of-week; 24h-lagged weather). Multi-step forecasting was recursive: each hour's prediction feeds back as a lag input for the next hour, so the backtest reflects a genuine 24h-ahead forecast rather than 24 independent one-step predictions. LightGBM outperformed XGBoost on every metric (MAE 233.2 vs. 251.7; RMSE 344.6 vs. 352.7; MASE 0.760 vs. 0.820); neither beat SARIMAX or the foundation model.

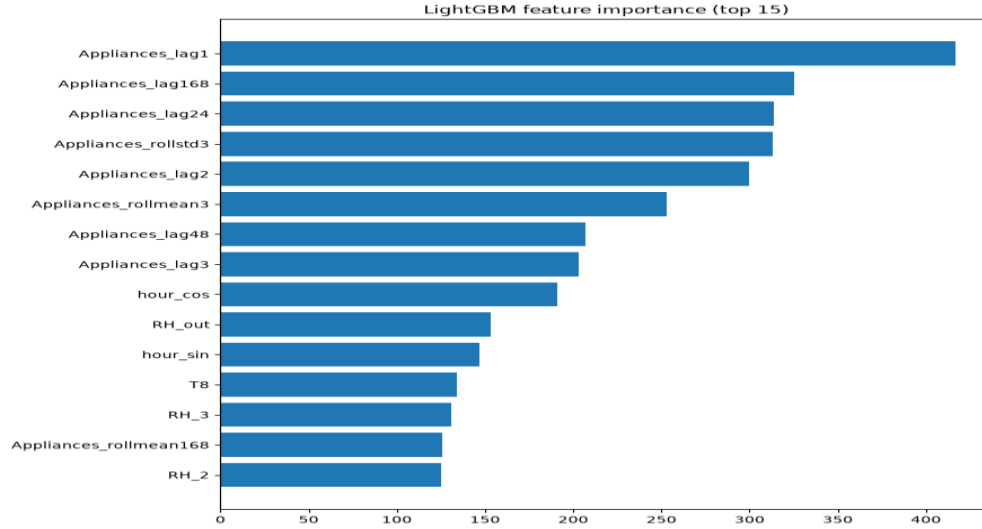


Figure 6. LightGBM feature importance (top 15).3

A feature-group ablation (LightGBM) isolates which groups drive accuracy:

Feature set	MAE	RMSE	MASE
Lags + rolling stats only	245.8	374.9	0.800
+ time-of-day / day-of-week	225.0	339.8	0.733
+ weather / indoor sensors (full)	238.6	351.2	0.777

Lags-plus-time is the strongest configuration; adding weather/sensor covariates on top makes MASE slightly worse (0.777) rather than better — a genuine negative result, not an artefact: an oracle (true future weather) version performed no better (MASE 0.83–0.84), confirming weather adds noise rather than signal at this feature granularity given the ~3,000-row training set.

8. Foundation Model (Chronos-Bolt-small)

Chronos-Bolt-small (Amazon) was applied purely zero-shot — no fine-tuning — taking the previous 28 days of Appliances as context and producing 10th/50th/90th-percentile quantile forecasts, on the identical 14-origin backtest.

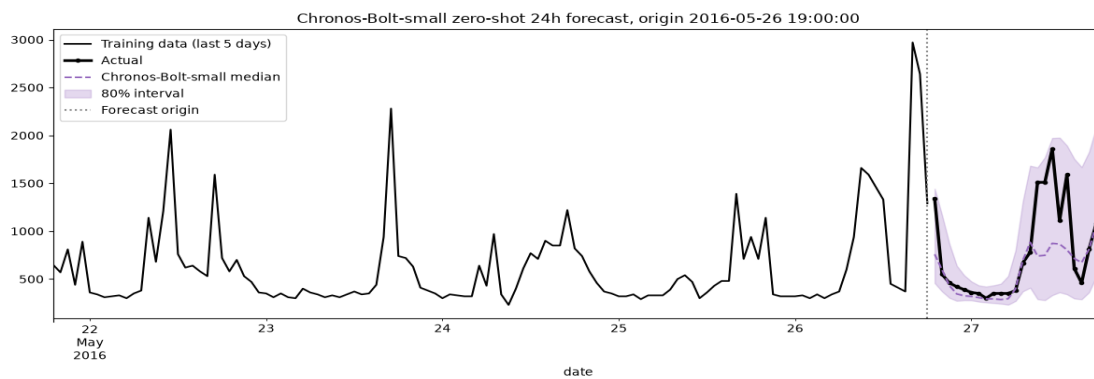


Figure 7. Chronos-Bolt-small zero-shot 24h forecast, 80% interval, final backtest origin.

Chronos-Bolt-small achieves MAE 202.4, RMSE 335.9, sMAPE 26.5%, and MASE 0.638 — the best MASE of any model, within 3% RMSE of SARIMAX, with no dataset-specific fitting at all. Its 80% intervals show 83.3% empirical coverage on average, reasonably well calibrated. Per-origin variance was wider than SARIMAX's: MASE ranged from 0.127 to 1.550 across the 14 origins, and coverage dropped to 62% on the hardest (20 May) origin — the same day that was hardest for every model, indicating a data property rather than a model-specific weakness.

9. Results

Model	MAE	RMSE	sMAPE	MASE
Chronos-Bolt-small (zero-shot)	202.4	335.9	26.5%	0.638
SARIMAX	208.3	325.8	29.0%	0.656
LightGBM	233.2	344.6	32.1%	0.760
XGBoost	251.7	352.7	35.1%	0.820
Mean (benchmark)	296.1	391.2	46.0%	0.932
Weekly seasonal-naive	254.6	404.9	32.5%	0.802
Daily seasonal-naive	288.6	455.8	35.5%	0.908
Naive (persistence)	512.0	584.7	65.4%	1.608
Drift	513.6	586.1	65.5%	1.613

Table 1. Consolidated 14-origin backtest results, sorted by MASE.

10. Discussion: Six Required Questions

Q1. Which benchmark is strongest, and what does this tell us about the structure of appliance use?

Weekly seasonal-naive is strongest (MASE 0.802), ahead of daily seasonal-naive (0.908) and the historical mean (0.932); plain naive and drift are far worse (1.608, 1.613). This shows appliance use follows a day-of-week routine — knowing what happened at this hour last week beats knowing what happened yesterday or an hour ago — so the dominant, exploitable structure is seasonality (daily and weekly), not trend or short-term momentum, matching the STL result in Section 3.

Q2. Does SARIMAX improve on the strongest seasonal benchmark? Are daily seasonality, autocorrelation, and exogenous variables adequately captured?

Yes: SARIMAX(0,0,6)(2,0,1,24) reaches MASE 0.656, an 18% improvement over weekly seasonal-naive (0.802), and the best RMSE of any model tested. Daily seasonality is captured well — the seasonal AR/MA terms at period 24 were chosen by AIC over simpler alternatives, a ~250-point improvement. Autocorrelation is not fully captured, however: Ljung-Box rejects the null of no residual autocorrelation at lags 24 and 48 ($p = 0.0018, 0.0023$), most plausibly the weekly effect, since the seasonal period was fixed at $m=24$ (not $m=168$) for computational tractability. No exogenous variables were included in this specification — it is a pure SARIMA model despite the “X” in its name. Section 7’s ablation suggests this is unlikely to cost much accuracy, since weather/sensor covariates did not help the tree-based models either once lag and time features were present, but this remains untested for SARIMAX specifically and is a natural extension.

Q3. Does the feature-based model improve with lag, rolling-window, time-of-day, and sensor/weather variables? Which feature groups are most useful?

Yes, substantially, but not uniformly. LightGBM’s ablation goes from MASE 0.800 (lags/rolling only) to 0.733 (+ time-of-day/day-of-week) — an 8% improvement — but adding weather/sensor covariates on top makes it worse again (0.777). Lag and rolling-window features are essential (dominating the correlation and feature-importance rankings, Figures 5–6); time-of-day/day-of-week features add real further value; weather and indoor-sensor features are not useful at this feature-engineering granularity — confirmed genuine, not an artefact of the leakage distinction, since even oracle (true future) weather did not help (MASE 0.83–0.84).

Q4. Does the foundation model outperform the benchmark, SARIMAX, and feature-based models? Is the improvement large enough to justify the complexity?

Against the benchmarks and ML models, yes and clearly: Chronos-Bolt-small’s MASE (0.638) is the best of any model, ahead of weekly seasonal-naive (0.802), LightGBM (0.760), and XGBoost (0.820). Against SARIMAX specifically, the two are effectively tied — Chronos leads narrowly on MASE (0.638 vs. 0.656) while

SARIMAX leads narrowly on RMSE (325.8 vs. 335.9). Given that near-tie, whether the complexity is justified depends on deployment scale: for a single house, Chronos's overhead (a pretrained neural model, ~191MB of weights, no interpretability) is not justified when SARIMAX matches it with a simple, transparent, fast-to-fit model. Across many households, though, Chronos's zero-shot property — one model, no per-house refitting — could justify the overhead, since SARIMAX would need a separate fit per house.

Q5. Which variables are genuinely known at the forecast origin? Is using future weather/sensor values a true or conditional forecast?

Only calendar/time features (hour, day-of-week) and the target's own past values are genuinely known 24 hours ahead. Indoor-sensor and weather readings at the forecast target hour are contemporaneous measurements, not forecasts — using their true values from the test set produces a conditional forecast (“if we already knew tomorrow's weather”), not a genuine one. This was tested directly: LightGBM with true future (oracle) weather reached RMSE 303.9, versus 345.4 with realistic (lagged) weather — a 13.7% inflation in apparent accuracy from the leakage, which is the measured cost of building a genuinely deployable forecast rather than a conditional one.

Q6. Considering accuracy, interpretability, uncertainty, computational cost, and deployment ease, which model is recommended?

Accuracy: SARIMAX and Chronos-Bolt-small are tied for best (MASE 0.656 / 0.638), both clearly ahead of the ML models and every benchmark. Interpretability: SARIMAX's explicit AR/MA coefficients and standard diagnostics are far more transparent than Chronos's neural weights or the tree ensembles' feature importances alone. Uncertainty: both SARIMAX and Chronos produce native, reasonably calibrated intervals (SARIMAX's Gaussian CI; Chronos's quantiles, 83.3% empirical coverage at nominal 80%); the ML models were not evaluated for calibrated intervals here. Computational cost: SARIMAX is far cheaper to fit and refit per house; Chronos needs no training at all but more compute per inference call and a large pretrained download. Deployment ease: SARIMAX is simplest to deploy and monitor for one house; Chronos's zero-shot property removes per-house tuning entirely, which matters once many households are involved. Recommendation: SARIMAX for a single-house deployment; Chronos-Bolt-small if the same pipeline must scale, zero-shot, across many households.

11. Limitations and Future Work

- Single house, single 4.5-month window (no summer/autumn data); a full year or multiple households would test generalisation.
- SARIMAX's order search used a 45-day subsample and a daily-only seasonal term ($m=24$), and included no exogenous variables (Q2); a weekly seasonal term or a SARIMAX-X extension could close some of the remaining gap.
- Chronos-Bolt-small's strong average masks high per-origin variance (MASE 0.127–1.550) and degraded interval coverage (62%) on the most volatile day — worth flagging for any deployment relying on its uncertainty estimates.
- Recursive multi-step forecasting inflates the ML models' error across the horizon by construction; a direct multi-horizon model could change the ML-vs-SARIMAX ranking.

12. Final Model Recommendation

As set out in Q6 above: SARIMAX is recommended for a single-house deployment, offering accuracy essentially tied with Chronos-Bolt-small alongside far greater interpretability and lower computational cost. Chronos-Bolt-small becomes preferable at multi-household scale, where its zero-shot property removes per-house tuning entirely. Weekly seasonal-naïve is a reasonable low-maintenance fallback if neither can be

maintained. The gradient-boosted ML models are not recommended: they cost more to build than SARIMAX and their extra covariates do not improve accuracy.

References

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